

Agus Fuad Mudhofar

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EDUCATION

Bachelor of Computer Engineering

Institut Teknologi Sepuluh Nopember, Surabaya

2022 - 2026

GPA: 3.64

EXPERIENCE

Research Intern – Mobile & Full-Stack Developer

Bandung

Feb - July 2026

Badan Riset dan Inovasi Nasional (BRIN)

- Designed and engineered a cross-platform Flutter application deployed for Mobile, Tablet, and Desktop environments to calculate agricultural nutrient requirements.
- Developed and optimized a RESTful API backend using Laravel 12 and PostgreSQL, implementing token-based authentication via Laravel Sanctum.
- Architected the application codebase as a monorepo to streamline multi-environment development, local testing, and build pipelines.
- Implemented automated workflows and clean backend/frontend separation to enhance code maintainability and deployment efficiency.

Front End Developer

Surabaya

Aug 2024 - Aug 2025

Multimedia and IoT Laboratory (MIOT)

- Designed and built the official MIOT laboratory web application using React.js and Tailwind CSS with a modular component architecture.
- Integrated dynamic content management using Git-managed JSON structures, allowing easy updates for research publications and team profiles.
- Configured continuous deployment pipelines using GitHub Pages to automate hosting and release processes.
- Optimized site performance, asset loading, and SEO configurations, resulting in improved search visibility and page load speeds.

Lab Teaching Assistant – Basic Programming & DMOJ Maintainer

Surabaya

Aug 2025 - Nov 2025

Multimedia and IoT Laboratory (MIOT)

- Responsible for deploying, configuring, and maintaining the DMOJ Online Judge system used by 100+ students for practical programming assessments.
- Configured and managed containerized applications using Docker and Docker Compose to ensure sandbox security and stable judge execution.
- Maintained the Django/Python-based web interface, diagnosing system logs and optimizing runtime resources under peak submission loads.
- Guided students in structured C/C++ programming, debugging syntax errors, and explaining algorithmic concepts.

Research Assistant – Multimedia and IoT Lab (B300)

Surabaya

April - June 2025

Multimedia and IoT Laboratory (MIOT)

- Conducted research on deploying deep learning models on edge devices using NVIDIA Jetson Nano, focusing on computer vision tasks.
- Developed and optimized object detection pipelines using Python and C++ with MobileNetSSD, YOLOv5n, and YOLOv11s architectures.
- Explored model conversion and acceleration techniques with ONNX and NVIDIA TensorRT for realtime inference.
- Integrated and evaluated models within the NVIDIA DeepStream SDK for efficient video analytics and deployment.
- Benchmarked performance and resource utilization to optimize inference speed and accuracy in constrained environments.

PROJECTS

Comic Reading Web Application (Full-Stack & DevOps)

A multi-architecture comic reading platform featuring optimized image rendering and automated pipelines.

- Developed and benchmarked dual backend implementations (Laravel-React and Express-React) to analyze performance in image-heavy rendering.
- Integrated MinIO as an S3-compatible local object storage solution, optimizing access speeds for gigabytes of self-hosted media.
- Configured Jenkins pipelines to automate the build, test, and deployment processes, establishing a continuous integration and delivery (CI/CD) workflow.
- Deployed the production-grade backend utilizing FrankenPHP to maximize concurrent request capacity and improve response latency.
- Technology Used: React, Laravel, Express.js, PostgreSQL, FrankenPHP, MinIO, Jenkins, Docker, Git.

ESAM: Enterprise Smart Automation & Monitoring System

A real-time containerized microservices dashboard for infrastructure telemetry, background worker execution, and AI-driven diagnostics.

- Architected a multi-service platform using React, C# .NET API, Spring Boot Java, and a Python AI Engine communicating via a RabbitMQ event bus.
- Integrated the Google Gemini API to analyze log streams and deliver structured anomaly diagnostic summaries and actionable troubleshooting recommendations.
- Engineered custom middleware to track live HTTP traffic stats (RPM, Latency) and dynamically monitor CPU, memory, and task queue depths.
- Configured a comprehensive GitHub Actions CI/CD pipeline featuring PostgreSQL and RabbitMQ container services to run JUnit verification tests and automate builds to GHCR.
- Technology Used: React, .NET 10, Spring Boot, Python, RabbitMQ, PostgreSQL, Docker, GitHub Actions, Google Gemini API.

IoT Developer – Smart Wheelchair Development

Aug 2024 - July 2026

Surabaya

Multimedia and IoT Laboratory (MIOT)

- Maintained and enhanced a smart wheelchair system utilizing deep learningbased computer vision for handsfree navigation and control.
- Integrated realtime eyegaze and facial gesture recognition using Python and OpenCV, improving system responsiveness and user accessibility.
- Collaborated with faculty and student teams to refine model architectures, optimize inference speed, and troubleshoot hardware/software integration issues.
- Developed custom microcontroller firmware (C/C++) for reliable motor actuation and sensor feedback, ensuring safe and accurate wheelchair movement.
- Documented system upgrades, conducted user testing, and presented results in lab meetings and academic showcases.

Camera Calibration & Image Processing Toolkit (Frontend & Python)

An interactive tool for computer vision experiments and camera calibration.

- Built a responsive single-page application using React, TypeScript, and Vite for interactive visualization of image processing stages.
- Developed image processing algorithms and camera calibration scripts in Python with OpenCV and NumPy.
- Technology Used: React, TypeScript, Vite, Python, OpenCV, NumPy.

Sappy: Multi-Role Livestock Management System

A mobile livestock management solution for multiple user roles featuring NFC tracking.

- Built a mobile app using Flutter and Dart supporting role-based dashboards (Farmer, Admin, Doctor) and multi-language localization.
- Implemented secure state management using Provider and integrated native NFC communication for physical asset tracking.
- Technology Used: Flutter, Dart, Provider, NFC, REST APIs.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, PHP, Python, SQL, C/C++, HTML/CSS, Dart

Frontend Web/Mobile: React.js, Next.js, Tailwind CSS, Vite, Flutter, HTML5, CSS3

Backend & Databases: Node.js (Express.js), Laravel, Django, PostgreSQL, MySQL

DevOps & Infrastructure: Jenkins, Docker, Docker Compose, MinIO, FrankenPHP, Git, GitHub Actions, Linux/Bash, Nginx, GitHub Pages

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Software Engineering, Computer Networks

Soft Skills: Problem Solving, Team Collaboration, Technical Documentation, Adaptability

Languages Proficiency: Indonesian (Native), English (B2 CEFR - Certified by DynEd)